

Sujay Seeram

Cyber-Physical Systems, Embedded Electronics & Physics-Informed Engineering

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ENGINEERING PROFILE & CORE PHILOSOPHY

Electronics and Communication Engineering undergraduate at Amrita Vishwa Vidyapeetham specializing in **resource-constrained cyber-physical systems, embedded hardware architecture, and physics-informed computing**. Pioneer of the “**Extreme Economic Engineering**” paradigm: deriving hardware configurations from physical first principles to drastically slash physical component counts, quiescent power consumption, and bill-of-materials (BOM) while maximizing system capability and sensor fidelity. Disciplined NCC cadet and 3-time Services Selection Board (SSB) attendee.

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
 - **Core Coursework:** Microprocessors & Microcontrollers (ARM Cortex/8051), Signals & Systems, Control Engineering, Digital System Design, Analog Electronic Circuits, VLSI Design, Electromagnetic Theory, Communication Networks, Power Electronics.
 - **Academic Focus:** Cyber-Physical Architectures, Acoustic/Vibration Signal Processing, Embedded Edge ML, Sensor Conditioning.

TECHNICAL SKILLS

- **Silicon, SoCs & MCUs:** STM32 (ARM Cortex-M4), ESP32 (Dual-Core Xtensa, RISC-V ULP), Raspberry Pi, RP2040, ATmega328P.
- **Firmware & Low-Level Software:** Embedded C, C++17/20, FreeRTOS, TinyML, Python, Assembly (ARM/x86), Linux Device Drivers.
- **Hardware Protocols & Buses:** SPI, I2C, UART/USART, CAN bus, PWM, ADC/DAC, DMA-driven waveform synthesis, GPIO edge interrupts.
- **Analog Conditioning & Sensors:** OPA1612 Butterworth active filters, ACS712 Hall-effect current sensors, ZMPT101B voltage transducers, MPU6050 6-DoF IMUs, ultrasonic/piezo transducers, capacitive discharge spot welders.
- **DSP, Signal Processing & Edge AI:** Fast Fourier Transform (FFT) spectral shift analysis, Mel-Frequency Cepstral Coefficients (MFCCs), Kalman filtering, Isolation Forest anomaly isolation, TFLite Micro.
- **Prototyping & Instrumentation:** KiCad PCB layout & schematic design, Digital Storage Oscilloscopes (DSOs), Logic Analyzers, JTAG/SWD.

LEADERSHIP, COMPETITIONS & PROFESSIONAL DISTINCTIONS

- **IIT Madras Smart India Challenge – 2nd Runner Up & Youngest Panelist** *IIT Madras*
 - Awarded **2nd Runner Up** at the prestigious IIT Madras Smart India Challenge for engineering economical cyber-physical solutions.
 - Honored as the **Youngest Panelist** at the summit, presenting on hardware miniaturization and extreme economic component engineering.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected delegate at the **GARC Chennai Meet**, engaging with automotive safety homologation, ADAS sensing, and powertrain testing.
 - Conducted technical discussions on non-invasive vehicle telemetry, EV battery safety envelopes, and real-time sensor failure preemption.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - Active cadet in the **National Cadet Corps (NCC)**, trained in field discipline, crisis leadership, obstacle navigation, and drill excellence.
 - Recommended for and attended the **Services Selection Board (SSB) 3 times**, demonstrating high situational awareness and psychological grit.

- **MotorSafe – Tri-Modal Edge-AI Predictive Maintenance Ecosystem**

 - Architected an edge-native predictive maintenance ecosystem monitoring industrial motors and preventing catastrophic bearing breakdowns.
 - Synchronized three high-frequency telemetry transducers: **ACS712** Hall-effect current (torque load), **ZMPT101B** (power quality), and **MPU6050** (vibrations).
 - Engineered on-chip Fast Fourier Transform (FFT) to extract spectral sidebands and deployed a localized unsupervised **Isolation Forest** on ESP32 to eliminate cloud latency and false alarms.
 - Streamed telemetry over MQTT into an interactive Three.js 3D digital twin dashboard reflecting motor health in real time.

Release v0.01
- **AQUAPULSE – Cognitive Software-Defined Sonar Payload for AUVs**

 - Engineered a cognitive, software-defined sonar payload for Autonomous Underwater Vehicles (AUVs) meeting the SIH26058 MoES/NIOT mandate.
 - Implemented zero-CPU DMA-to-DAC acoustic waveform synthesis on **STM32 / ESP32-S3** paired with active **OPA1612** Butterworth analog filtering.
 - Deployed an INT8 quantized **TinyML** edge policy engine to dynamically adapt ping frequency (100–480 kHz) across shifting thermoclines, cutting acoustic power consumption by **38%**.

Repository
- **EVolvAI – Physics-Informed EV Charging Infrastructure Engine**

 - Formulated a research-grade cyber-physical framework coupling a **Physics-Constrained Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm (GA).
 - Simulated distribution grid bus voltage drops, transformer thermal constraints, and climate stress anomalies to optimize city-scale charging node placement, achieving $R^2 \approx 0.5$ fidelity on sparse real-world data.

Release v1.0.0
- **SpecRAG – Zero-Hallucination Firmware & Datasheet Auditing Gateway**

 - Engineered an enterprise developer tool bridging 1,000-page silicon hardware datasheets with embedded C/C++ firmware.
 - Utilized the **Model Context Protocol (MCP)** and localized vector search to verify register bitmasks, clock-tree prescalers, and alternate-function pin muxing directly in the IDE prior to board fabrication.

Release v0.1.0
- **Pheno-Speak – Edge Acoustic Vocal Biomarker Extraction Engine**

 - Built an edge audio digital signal processing pipeline for real-time vocal biomarker extraction aligned with SDG 3.4 (Digital Phenotyping).
 - Extracted MFCCs, jitter, and shimmer features on lightweight edge hardware without offloading unencrypted audio to cloud endpoints.

Repository
- **SAFE-Inventory – Low-Power BLE Warehouse Asset Tracking Suite**

 - Engineered a resilient warehouse asset management system featuring single-product batch scanning and set-and-forget Bluetooth LE hardware pairing.
 - Optimized transceiver duty-cycling on mobile and embedded scanners to provide multi-shift operational battery endurance.

Release v1.4.0
- **Hardware-Qwen – Quantized SLM Inference on Resource-Constrained Boards**

 - Optimized quantized Small Language Models (SLMs) for execution on embedded Linux SBCs and neural accelerator boards.
 - Implemented custom INT4/INT8 quantization kernels and memory-mapped model streaming to fit within constrained 1GB/2GB DDR RAM budgets.

Repository
- **Dust-Vision & Dust-Sim – Edge Particulate Pollution Tracking**

 - Engineered computer vision and optical particulate tracking models on edge cameras to quantify real-time construction site dust emissions.
 - Coupled sensor data with a custom particle kinematics physics simulation (**dust-sim v1.0.0**) modeling dispersion under thermal/wind shifts.

Release v0.1